

Практикум E9:archive

Twenty-nine questions, 147 marks. Every question the archive asks about how fast things change and where they are best, from May 2021 to November 2025, in the order of the nine techniques rather than the order of the papers.

О практикуме

calculus.optimization (18 блоков, 60 баллов) и calculus.rates_of_change (14 блоков, 56) целиком и десять блоков calculus.differentiation на 31 балл, отданных сюда при разборе E3: $v = ds/dt$, $a = dv/dt$, скорость роста растения. Всего 42 блока и 147 баллов; check_map.py показывает 41, потому что десять блоков он считает долей. Последний практикум карты.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e9-optimisation-rates

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

E9 archive — optimisation and related rates, all of them

Twenty-nine questions, 147 marks.

Every question the archive asks about how fast things change and where they are best, from May 2021 to November 2025, in the order of the nine techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-e9-optimisation-rates.ipynb`.

§	technique	parts	marks
1	A rate at a moment	8	19
2	Rest and turning	6	17
3	The largest value in motion	4	14
4	A chain through one quantity	4	23
5	A chain through an angle	3	20
6	The optimum of a model	5	16
7	The shortest distance	3	14
8	The fastest change	4	16
9	The best whole number	3	8

The checks are the same ones the practicum uses and they store nothing. A particle is handed over as the question gives it — `particle(v=...)` or `particle(s=...)` — and the check **measures** its acceleration by moving time a little; a related rate is the one that keeps the relation true when the picture moves; a best value is found by looking at the whole interval, its ends, and — when the letter counts things — every whole number. Three significant figures, unless the question asks otherwise.

Parts of one question are split by technique. The particle of May 2024 TZ2 appears in §§ 1, 2 and 3; the logistic model of May 2022 TZ2 in §§ 1 and 8. Each part repeats what it needs.

Two parts are twins. November 2023 Paper 2 was set as TZ1 and TZ2 and is one paper. Its Q4(a) and Q8(b) stand here as the TZ2 copy; the TZ1 copies are in the E8 and C7 archives.

Two answers are a choice between phrasings, 'a' to 'd': there is nothing to compute in them, and the check knows only a digest.

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/calculus to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
particle, rate_of, dt

language('en')               # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. A rate at a moment

Eight parts, 19 marks. Name whose rate it is, then take the derivative at the moment — or solve for the moment when the rate is given.

1.1 — May 2024 TZ1 Paper 2 Q10(c), 2 marks

The height of water at Sule Skerry, in metres, is $H(t) = 1.63 \sin(0.513(t - 8.20)) + 2.13$, where t is the number of hours after midnight. Find the rate of change of the height of the water when $t = 13$, in metres per hour.

```
q1_1 = ...          # H'(13)
```

```
verify_rate('1.1', q1_1, 1.63*sin(0.513*(t -  
8.20)) + 2.13, 13)
```

1.2 — May 2021 TZ1 Paper 2 Q4(c), 2 marks

A particle moves in a straight line with velocity $v(t) = t \sin t - 3 \text{ m s}^{-1}$ for $0 \leq t \leq 10$. Find the acceleration of the particle when $t = 7$.

```
q1_2 = ...          # the acceleration at t = 7

path = particle(v=t*sin(t) - 3, span=(0, 10))

verify_motion('1.2', q1_2, path,
              'acceleration', 7)
```

1.3 — May 2022 TZ2 Paper 2 Q5(b), 3 marks

A particle moves in a straight line with velocity $v = \frac{(t^2 + 1) \cos t}{4} \text{ m s}^{-1}$, $0 \leq t \leq 3$. Find the times when the particle's acceleration is -1.9 m s^{-2} .

Both times, as a list.

```
q1_3 = [...]        # every time when a = -1.9

path = particle(v=(t**2 + 1)*cos(t)/4, span=
              (0, 3))

verify_when('1.3', q1_3, path,
            ('acceleration', -1.9), 'all')
```

1.4 — May 2025 TZ3 Paper 2 Q10(b), 2 marks

Fiona's velocity during a race is $v(t) = \frac{8.14t}{\sqrt{t^2 + 0.2}} \text{ m s}^{-1}$, $t \geq 0$. Find the time when Fiona's acceleration is 4 m s^{-2} .

```

q1_4 = ...          # the time when a = 4

path = particle(v=8.14*t/sqrt(t**2 + 0.2),
span=(0, oo))

verify_when('1.4', q1_4, path,
('acceleration', 4))

```

1.5 — May 2023 TZ2 Paper 2 Q5(a), 2 marks

A particle moves in a straight line with velocity $v(t) = 4e^{-t/3} \cos\left(\frac{t}{2} - \frac{\pi}{4}\right)$ m s⁻¹ for $0 \leq t \leq 4\pi$. Let t_1 be the first time when the particle's **acceleration** is zero. Find t_1 .

```

q1_5 = ...          # t_1

path = particle(v=4*exp(-t/3)*cos(t/2 -
pi/4), span=(0, 4*pi))

verify_when('1.5', q1_5, path,
('acceleration', 0))

```

1.6 — May 2022 TZ2 Paper 2 Q10(d), 6 marks

The height of Plant A is $h_A(t) = \sin(2t + 6) + 9t + 27$ cm and of Plant B $h_B(t) = 8t + 32$ cm, t in weeks, $0 \leq t \leq 9$. Find the total

amount of time when the rate of growth of Plant B was greater than the rate of growth of Plant A.

```
q1_6 = ...          # total time, weeks

verify_duration('1.6', q1_6, 8*t + 32,
sin(2*t + 6) + 9*t + 27, (0, 9))
```

1.7 — May 2024 TZ2 Paper 2 Q4(a), 1 mark

A particle moves in a straight line with velocity $v(t) = 1 + e^{-t} - e^{-\sin 2t}$ m s⁻¹ for $0 \leq t \leq 2$. Find the velocity of the particle at $t = 2$.

```
q1_7 = ...          # v(2)

path = particle(v=1 + exp(-t) - exp(-
sin(2*t)), span=(0, 2))

verify_motion('1.7', q1_7, path, 'velocity',
2)
```

1.8 — May 2022 TZ2 Paper 2 Q12(a), 1 mark

The population P of a species of marsupial is modelled by $\frac{dP}{dt} = kP \left(1 - \frac{P}{N}\right)$, where t is the time in years.

In the context of the model, interpret the meaning of $\frac{dP}{dt}$.

- 'a' — the population of marsupials at time t
- 'b' — the rate of change of the population with respect to time
- 'c' — the growth in the population per year
- 'd' — the largest population the island can sustain

```
q1_8 = ...          # 'a', 'b', 'c' or 'd'  
check_word('1.8', q1_8, '3e23e8160039')
```

§ 2. Rest and turning

Six parts, 17 marks. At rest: $v = 0$.
Changes direction: v changes sign.
Displacement increasing: $v > 0$. Only the roots inside the interval count, and only as many as are asked.

2.1 — *May 2021 TZ1 Paper 2 Q4(a), 2 marks*

A particle moves in a straight line with velocity $v(t) = t \sin t - 3 \text{ m s}^{-1}$ for $0 \leq t \leq 10$. Find the smallest value of t for which the particle is at rest.

```
q2_1 = ...          # the first time at rest

path = particle(v=t*sin(t) - 3, span=(0, 10))

verify_when('2.1', q2_1, path, 'rest')
```

2.2 — *November 2023 TZ2 Paper 2 Q4(a), 2 marks*

A particle moves along a straight line. Its displacement from a fixed point O after t seconds is $s(t) = 4.3 \sin(\sqrt{3t+5}) \text{ m}$, $0 \leq t \leq 10$. The particle first comes to rest after q seconds. Find q .

The TZ1 copy of this part is in the E8 archive.

```
q2_2 = ...          # q

path = particle(s=4.3*sin(sqrt(3*t + 5)),
span=(0, 10))

verify_when('2.2', q2_2, path, 'rest')
```

2.3 — May 2024 TZ1 Paper 2 Q4(a) and (b), 4 marks

A particle moves in a straight line with velocity $v = 2 \sin(0.5t) + 0.3t - 2 \text{ m s}^{-1}$ for $0 \leq t \leq 10$.

(a) Find the smallest value of t when the particle changes direction.

(b) Find the range of values of t for which the displacement of the particle is increasing. As *Interval(a, b)*.

```

q2_3a = ...      # the first change of
direction
q2_3b = ...      # when the displacement
increases

path = particle(v=2*sin(0.5*t) + 0.3*t - 2,
span=(0, 10))

verify_when('2.3(a)', q2_3a, path, 'turn')
verify_while('2.3(b)', q2_3b, path,
'forward')

```

2.4 — May 2022 TZ1 Paper 2 Q4(b), 3 marks

A particle moves along a straight line with velocity $v(t) = e^{\sin t} + 4\sin t$ m s⁻¹ for $0 \leq t \leq 6$. Find the acceleration of the particle when it changes direction.

```

q2_4 = ...      # the acceleration at the
change of direction

path = particle(v=exp(sin(t)) + 4*sin(t),
span=(0, 6))

verify_motion('2.4', q2_4, path,
'acceleration', 'turn')

```

2.5 — May 2024 TZ2 Paper 2 Q4(c), 3 marks

A particle moves in a straight line with velocity $v(t) = 1 + e^{-t} - e^{-\sin 2t}$ m s⁻¹ for $0 \leq t \leq 2$. Find the acceleration of the particle at the instant it changes direction.

```
q2_5 = ...          # the acceleration at the
change of direction

path = particle(v=1 + exp(-t) - exp(-
sin(2*t)), span=(0, 2))

verify_motion('2.5', q2_5, path,
'acceleration', 'turn')
```

2.6 — May 2025 TZ2 Paper 2 Q5(c), 3 marks

A particle P moves in a straight line with velocity $v(t) = e^{-\sin t} \cos(2t)$ m s⁻¹ for $0 \leq t \leq 5$. Find the acceleration when P changes direction for the second time.

```
q2_6 = ...          # the acceleration at the  
second change  
  
path = particle(v=exp(-sin(t))*cos(2*t),  
span=(0, 5))  
  
verify_motion('2.6', q2_6, path,  
'acceleration', ('turn', 2))
```

§ 3. The largest value in motion

Four questions, 14 marks. Compare the vertices **and** the ends of the interval. Speed is $|v|$.

3.1 — May 2022 TZ2 Paper 2 Q5(c), 2 marks

A particle moves in a straight line with velocity $v = \frac{(t^2 + 1)\cos t}{4}$ m s⁻¹, $0 \leq t \leq 3$. Find the particle's acceleration when its speed is at its greatest.

```
q3_1 = ...          # the acceleration when the
speed is greatest

path = particle(v=(t**2 + 1)*cos(t)/4, span=
(0, 3))

verify_motion('3.1', q3_1, path,
'acceleration', 'fastest')
```

3.2 — May 2024 TZ2 Paper 2 Q4(b), 2 marks

A particle moves in a straight line with velocity $v(t) = 1 + e^{-t} - e^{-\sin 2t}$ m s⁻¹ for

$0 \leq t \leq 2$. Find the maximum velocity of the particle.

```
q3_2 = ...          # the maximum velocity

path = particle(v=1 + exp(-t) - exp(-
sin(2*t)), span=(0, 2))

verify_extreme('3.2', q3_2, path, 'velocity')
```

3.3 — May 2025 TZ2 Paper 2 Q5(a), 2 marks

A particle P moves in a straight line with velocity $v(t) = e^{-\sin t} \cos(2t)$ m s⁻¹ for $0 \leq t \leq 5$. Find the maximum speed of P.

```
q3_3 = ...          # the maximum speed

path = particle(v=exp(-sin(t))*cos(2*t),
span=(0, 5))

verify_extreme('3.3', q3_3, path, 'speed')
```

3.4 — November 2025 TZ3 Paper 2 Q10(a) and (b), 8 marks

A particle P moves in a straight line so that its displacement from a fixed point O at time t seconds is $s(t) = 2^{(1-\frac{t}{5})} \sin\left(\frac{2\pi t}{3}\right)$ cm, $t \geq 0$.

(a) Find (i) the maximum displacement of P from O; (ii) the maximum velocity of P.

(b) Find (i) the minimum value of the displacement function $s(t)$; (ii) the displacement of P from O when $t = 3.5$.

```
q3_4a = ...      # the maximum displacement
q3_4b = ...      # the maximum velocity
q3_4c = ...      # the minimum displacement
q3_4d = ...      # s(3.5)

path = particle(s=2*(1 - t/5)*sin(2*pi*t/3),
span=(0, oo))

verify_extreme('3.4(a)(i)', q3_4a, path,
'displacement')
verify_extreme('3.4(a)(ii)', q3_4b, path,
'veLOCITY')
verify_extreme('3.4(b)(i)', q3_4c, path,
'displacement', 'min')
verify_motion('3.4(b)(ii)', q3_4d, path,
'displacement', 3.5)
```

§ 4. A chain through one quantity

Four questions, 23 marks. The relation first, then the chain, then the state of the moment, then the numbers.

4.1 — May 2023 TZ1 Paper 1 Q6, 5 marks

The side lengths, x cm, of an equilateral triangle are increasing at a rate of 4 cm s^{-1} . Find the rate at which the area of the triangle, $A \text{ cm}^2$, is increasing when the side lengths are $5\sqrt{3}$ cm.

No calculator: exact.

```
q4_1 = ...          # dA/dt

A = Symbol('A')
triangle = measure('triangle', (0, 0, 0), (x,
0, 0), (x/2, sqrt(3)*x/2, 0))

verify_related('4.1', q4_1, Eq(A, triangle),
Eq(x, 5*sqrt(3)), {dt(x): 4}, dt(A),
                  where={x: (0, 20)},
exact=True)
```

4.2 — May 2024 TZ2 Paper 2 Q6, 6 marks

The volume of a spherical bubble increases at a constant rate of $5 \text{ cm}^3 \text{ s}^{-1}$; its initial volume can be taken as zero. Find the rate, in cm s^{-1} , at which the radius of the bubble is increasing when the volume of the bubble is 20 cm^3 .

```
q4_2 = ...          # dr/dt when V = 20

V, r = symbols('V r')

verify_related('4.2', q4_2, Eq(V,
4*pi*r**3/3), Eq(V, 20), {dt(V): 5}, dt(r),
               where={r: (0, 10)})
```

4.3 — May 2022 TZ1 Paper 2 Q10(e), 6 marks

A water container is modelled by rotating $y = \sqrt{x^2 - 1}$, $1 \leq x \leq 2$, about the y -axis. When it is filled to a height of h metres, the volume of water is $V = \pi \left(\frac{1}{3}h^3 + h \right) \text{ m}^3$, and the container is full at $h = \sqrt{3}$. Water is added at a constant rate of $0.4 \text{ m}^3 \text{ s}^{-1}$.

Find the rate of change of the height of the water when the container is filled to half its maximum volume.

```
q4_3 = ...          # dh/dt at half the maximum
volume

V, h = symbols('V h')
full = pi*(sqrt(3)**3/3 + sqrt(3))

verify_related('4.3', q4_3, Eq(V, pi*(h**3/3
+ h)), Eq(V, full/2), {dt(V): 0.4}, dt(h),
                    where={h: (0, sqrt(3))})
```

4.4 — November 2022 Paper 2 Q8, 6 marks

Liquid is poured into a round-bottomed flask at $2 \text{ cm}^3 \text{ s}^{-1}$. The volume $V \text{ cm}^3$ and height $h \text{ cm}$ of the liquid satisfy $V = 5\pi h^2 - \frac{1}{3}\pi h^3$, and the liquid never reaches the neck: $0 \leq h \leq 10$. Find the rate of change of the height of the liquid when the volume is 200 cm^3 .

```
q4_4 = ...          # dh/dt when V = 200

V, h = symbols('V h')

verify_related('4.4', q4_4, Eq(V, 5*pi*h**2 -
pi*h**3/3), Eq(V, 200), {dt(V): 2}, dt(h),
                where={h: (0, 10)})
```

§ 5. A chain through an angle

Three questions, 20 marks. The picture gives the relation; the second condition gives the state; *speed* has no sign, *how fast is it changing* does.

5.1 — May 2021 TZ1 Paper 2 Q9(b), 6 marks

Two boats A and B travel due north; initially B is 50 m due east of A. After t seconds A has travelled x m and B y m, and θ is the bearing of B from A. Part (a) shows $y = x + 50 \cot \theta$. At time T boat B has travelled 10 m further than A, B is travelling at double the speed of A, and θ changes at -0.1 rad s^{-1} . Find the speed of boat A at time T .

```

q5_1 = ...          # the speed of boat A

y, theta = symbols('y theta')

verify_related('5.1', q5_1, Eq(y, x +
50*cot(theta)), Eq(y - x, 10),
               [Eq(dt(y), 2*dt(x)),
Eq(dt(theta), -0.1)], dt(x),
               where={theta: (0, pi/2)},
size=True)

```

5.2 — May 2025 TZ3 Paper 2 Q9(b), 7 marks

An airplane flies at a constant height of 6 km towards a runway AB of length 2 km. With $GA = x$ km, the viewing angle of the runway is $\theta = \arctan\left(\frac{x+2}{6}\right) - \arctan\left(\frac{x}{6}\right)$. When the viewing angle is 0.178 radians it is changing at 12.5 radians per hour. Find the speed of the airplane.

```

q5_2 = ...          # the speed, km per hour

theta = Symbol('theta')

verify_related('5.2', q5_2, Eq(theta, atan((x
+ 2)/6) - atan(x/6)), Eq(theta, 0.178),
               {dt(theta): 12.5}, dt(x),
where={x: (0, 50)}, size=True)

```

5.3 — November 2025 TZ1 Paper 2

Q9, 7 marks

Triangle ABC has $AB = 15$ cm, $BC = 20$ cm, $AC = x$ cm and $\hat{ABC} = \theta$, $0 < \theta < \frac{\pi}{2}$. Angle θ is decreasing at $\frac{\pi}{60}$ radians per minute. Determine how fast the length of AC is changing when the area of the triangle is 140 cm^2 .

```
q5_3 = ...          # dx/dt, with its sign

S, theta = symbols('S theta')
B_, C_, A_ = (0, 0, 0), (20, 0, 0),
(15*cos(theta), 15*sin(theta), 0)

verify_related('5.3', q5_3, [Eq(x,
distance(A_, C_)), Eq(S, measure('triangle',
A_, B_, C_))],
               Eq(S, 140), {dt(theta): -
pi/60}, dt(x), where={theta: (0, pi/2)})
```

§ 6. The optimum of a model

Five parts, 16 marks. $f' = 0$ inside the domain, a reason for the kind, and then the thing actually asked.

6.1 — May 2023 TZ2 Paper 1 Q10(d), 6 marks

Triangle PQR is inscribed in $x^2 + y^2 = 9$ with $P(-3, 0)$, $Q(x, y)$ in the first quadrant and $R(x, -y)$ in the fourth. Its area is $A = (x + 3)\sqrt{9 - x^2}$, and $\frac{dA}{dx} = \frac{9 - 3x - 2x^2}{\sqrt{9 - x^2}}$. Find the y -coordinate of R such that A is a maximum.

No calculator: exact.

```
q6_1 = ...          # the y-coordinate of R

verify_best('6.1', q6_1, (x + 3)*sqrt(9 -
x**2), Interval.open(0, 3), 'max',
              report=-sqrt(9 - x**2),
exact=True)
```

6.2 — November 2025 TZ3 Paper 1 Q8(c), 4 marks

Astrid's time to reach Bronwyn is $T = 500 \sec \theta + \frac{2500 - 1000 \tan \theta}{3}$ seconds, $0 < \tan \theta < \frac{5}{2}$, and she chooses θ to make T as small as possible. The paper asks to show that then $PX = 400 \tan \theta = 160\sqrt{5}$ m.

Enter $\sin \theta$ at the quickest route — the line of working that carries the marks.

No calculator: exact.

```
q6_2 = ...          # sin(theta) at the minimum
of T

theta = Symbol('theta')
T_walk = 500*sec(theta) + (2500 -
1000*tan(theta))/3

verify_best('6.2', q6_2, T_walk,
Interval.open(0, atan(Rational(5, 2))),
'min',
                var=theta, report=sin(theta),
exact=True)
```

6.3 — May 2023 TZ1 Paper 1 Q11(d), 3 marks

A segment AB through the corner C of a passage has length $L = \frac{3}{4} \sec \alpha + 6 \operatorname{cosec} \alpha$,

$0 < \alpha < \frac{\pi}{2}$, and $\frac{dL}{d\alpha} = 0$ at $\alpha = \arctan 2$.

(i) Justify that L is a minimum there: which way is the graph of L concave at $\alpha = \arctan 2$? Enter 'up' or 'down'.

(ii) Determine this minimum value of L .
Exact.

```
q6_3a = ...      # 'up' or 'down'
q6_3b = ...      # the minimum of L

alpha = Symbol('alpha')
L = Rational(3, 4)*sec(alpha) +
6*cosec(alpha)

verify_concavity('6.3(i)', q6_3a, L, atan(2),
var=alpha)
verify_best('6.3(ii)', q6_3b, L,
Interval.open(0, pi/2), 'min', var=alpha,
exact=True)
```

6.4 — May 2023 TZ1 Paper 1 Q11(e), 2 marks

Two people need to carry a pole of length 11.25 m horizontally from the passageway of 6.3 into the room. Determine whether this is possible. Enter 'yes' or 'no'.

```

q6_4 = ...          # 'yes' or 'no'

alpha = Symbol('alpha')

verify_fits('6.4', q6_4, Rational(3,
4)*sec(alpha) + 6*cosec(alpha),
            Interval.open(0, pi/2), 11.25,
var=alpha)

```

6.5 — May 2025 TZ3 Paper 3 Q2(e), 1 mark

An open box is folded from an $a \times b$ sheet of cardboard: $V = x(a - 2x)(b - 2x)$, $0 \leq x \leq \frac{a}{2}$. Part (a) gives $V = 0$ at $x = 0$ and at $x = \frac{a}{2}$, and part (d) shows a local maximum of V at $x = x_m$ inside the interval. Explain why x_m gives the maximum volume of the box.

- 'a' — V is zero at both ends of the interval and positive at x_m , so the local maximum is the largest value
- 'b' — because $\frac{dV}{dx} = 0$ at x_m
- 'c' — because $\frac{d^2V}{dx^2} < 0$ at x_m
- 'd' — because x_m is the smaller solution of $\frac{dV}{dx} = 0$

```
q6_5 = ...          # 'a', 'b', 'c' or 'd'  
check_word('6.5', q6_5, 'ca978112ca1b')
```

§ 7. The shortest distance

Three questions, 14 marks. Minimise the square; one parameter for two moving points; the root only at the end.

7.1 — May 2025 TZ2 Paper 1 Q6(b), 6 marks

$f(x) = \sqrt{x^2 \ln x + 4 - x^2}$, $x > 0$. The distance from the origin to a point of the graph is $l = \sqrt{x^2 \ln x + 4}$. Find the x -coordinate of the point on the graph of f which is closest to the origin.

No calculator: exact.

```
q7_1 = ...          # x of the closest point

verify_best('7.1', q7_1, sqrt(x**2*log(x) +
4), Interval.open(0, 10), 'min',
              report='place', exact=True)
```

7.2 — May 2022 TZ2 Paper 2 Q11(e), 5 marks

Two airplanes have position vectors, in km, after t minutes, $\mathbf{r}_A = \begin{pmatrix} 19 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} -6 \\ 2 \\ 4 \end{pmatrix}$

and $\mathbf{r}_B = \begin{pmatrix} 1 \\ 0 \\ 12 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$, $0 \leq t \leq 2.5$. Let $D(t)$ be the distance between them. Find the minimum value of $D(t)$.

```
q7_2 = ...          # the minimum of D(t)

r_A = Matrix([19, -1, 1]) + t*Matrix([-6, 2,
4])
r_B = Matrix([1, 0, 12]) + t*Matrix([4, 2,
-2])

verify_best('7.2', q7_2, distance(r_A, r_B),
(0, 2.5), 'min', var=t)
```

7.3 — November 2023 TZ2 Paper 2 Q8(b), 3 marks

For the points $A(0, p, 2)$, $B(1, 1, 1)$, $C(p, 0, 4)$, $p > 0$, part (a) shows

$$\vec{AB} \times \vec{AC} = \begin{pmatrix} 2 - 3p \\ -2 - p \\ p^2 - 2p \end{pmatrix}$$

Find the smallest possible value of $|\vec{AB} \times \vec{AC}|^2$.

The TZ1 copy of this part is in the C7 archive.

```
q7_3 = ...          # the smallest |AB x AC|^2

p = Symbol('p')
product = Matrix([2 - 3*p, -2 - p, p**2 -
2*p])

verify_best('7.3', q7_3,
product.dot(product), Interval.open(0, 20),
'min', var=p)
```

§ 8. The fastest change

Four parts, 16 marks. The quantity made largest is itself a rate: its vertex is where the second derivative vanishes.

8.1 — May 2021 TZ1 Paper 2 Q12(e), 3 marks

The population of a colony of ants is $P = \frac{1200k}{1200 + (k - 1200)e^{-t/5}}$, $t \geq 0$ in days, and part (d) gives $k = 2845.35$ (2845.347...). Find the value of t when the rate of change of the population is at its maximum.

```
q8_1 = ...          # t at the fastest growth
```

```
k_ants = 2845.347  
ants = 1200*k_ants/(1200 + (k_ants -  
1200)*exp(-t/5))
```

```
verify_best('8.1', q8_1, rate_of(ants), (0,  
40), 'max', var=t, report='place')
```

8.2 — May 2021 TZ2 Paper 2 Q11(d), 6 marks

A bowl is formed by rotating $y = f(x) = \frac{ke^{x/2}}{1 + e^x}$, $0 \leq x \leq \ln 16$, about the x -axis, and part (b) gives $k = \sqrt{\frac{680}{\pi}}$. $f(x)$ is the cross-sectional radius at distance x .

(i) By sketching the graph of a suitable derivative of f , find where the cross-sectional radius of the bowl is decreasing most rapidly.

(ii) State the cross-sectional radius at this point.

```
q8_2a = ...      # x where the radius falls
fastest
q8_2b = ...      # the radius there

k_bowl = sqrt(680/pi)
radius = k_bowl*exp(x/2)/(1 + exp(x))

verify_best('8.2(i)', q8_2a, rate_of(radius,
x), (0, log(16)), 'min', report='place')
verify_best('8.2(ii)', q8_2b, rate_of(radius,
x), (0, log(16)), 'min', report=radius)
```

8.3 — May 2022 TZ2 Paper 2 Q12(c) and (d), 7 marks

The population P of a species is modelled by $\frac{dP}{dt} = kP \left(1 - \frac{P}{N}\right)$, with k, N positive and $0 < P < N$.

(c) Find the value of P at which the population increases at its maximum rate. (The paper says: show that it is $\frac{N}{2}$, and justify.)

(d) Hence determine the maximum value of $\frac{dP}{dt}$ in terms of k and N .

```
q8_3 = ...           # the population at the
maximum rate
q8_4 = ...           # the maximum of dP/dt

pop = Symbol('pop')      # the
population of the question
growth = k*pop*(1 - pop/N)

verify_best('8.3', q8_3, growth, (0, N),
'max', var=pop, report='place',
           params={k: (0.7, 0.3, 2), N:
(100, 7, 50)})
verify_best('8.4', q8_4, growth, (0, N),
'max', var=pop,
           params={k: (0.7, 0.3, 2), N:
(100, 7, 50)})
```

§ 9. The best whole number

Three parts, 8 marks. $M_n(S) = \left(\frac{S}{n}\right)^n$ is the maximum product of n positive numbers with sum S , and $P(S)$ is its largest value over whole n . Compare the whole numbers either side of the vertex.

9.1-9.3 — May 2023 TZ1 Paper 3 **Q2(f), (g), (j), 8 marks**

9.1 Write down the value of $P(12)$ and the value of n at which it occurs.

9.2 Determine the value of $P(20)$ and the value of n at which it occurs.

9.3 Find the largest possible product of positive numbers whose sum is 100, in the form $a \times 10^k$ with $1 \leq a < 10$.

Pairs as (P, n) .

```
q9_1 = ...      # (P(12), n)
q9_2 = ...      # (P(20), n)
q9_3 = ...      # the largest product with
sum 100

n = Symbol('n')

verify_best('9.1', q9_1, (12/n)**n, (1, 40),
'max', var=n, report=('value', 'place'),
integer=True)
verify_best('9.2', q9_2, (20/n)**n, (1, 40),
'max', var=n, report=('value', 'place'),
integer=True)
verify_best('9.3', q9_3, (100/n)**n, (1,
200), 'max', var=n, integer=True)
```



Solutions

§ 1

1.1 $H'(13) = \boxed{-0.651}$ m per hour
($-0.650622\dots$).

1.2 $a = v'(7) = \sin 7 + 7 \cos 7 = \boxed{5.93}$ m s⁻²
($5.93430\dots$).

1.3 $a(t) = v'(t) = -1.9$ on $[0, 3]$ at
 $\boxed{t = 2.26, 2.96}$ s.

1.4 Solving $v'(t) = 4$: $\boxed{t = 0.591}$ s.

1.5 $v'(t) = 0$ first at the local maximum of

v : $\boxed{t_1 = \arctan \frac{5}{12} \approx 0.395}$ s.

1.6 $h'_B > h'_A$ means $\cos(2t + 6) < -\frac{1}{2}$: three
stretches of length $\frac{\pi}{3}$, total $\boxed{\pi \approx 3.14}$
weeks.

1.7 $v(2) = 1 + e^{-2} - e^{-\sin 4} = \boxed{-0.996}$ m s⁻¹.

1.8 $\boxed{b}$ — the rate of change of the
population with respect to time. "*Growth
per year*" is explicitly not accepted.

§ 2

2.1 $t \sin t = 3$ first at $t = 6.74$ s; 9.09 is the second root and is not asked.

2.2 $v = \frac{ds}{dt} = \frac{3 \cdot 4.3 \cos \sqrt{3t+5}}{2\sqrt{3t+5}} = 0$ when $\sqrt{3t+5} = \frac{3\pi}{2}$ (the first such value above $\sqrt{5}$), so $t = 5.74$.

2.3 (a) $t = 1.69$; **(b)** $v > 0$ between the roots: $1.69 < t < 6.12$.

2.4 $v = 0$ at $t = 3.34692 \dots$; $a = v'(3.346 \dots) = -4.71 \text{ m s}^{-2}$.

2.5 $v = 0$ at $t = 1.65840 \dots$; $a = -2.53 \text{ m s}^{-2}$.

2.6 $v = 0$ at $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$; at the second, $a = 2e^{-1/\sqrt{2}} = 0.986 \text{ m s}^{-2}$.

§ 3

3.1 $|v|$ is greatest at the end $t = 3$; $a = v'(3) = -1.84 \text{ m s}^{-2}$.

3.2 $v'(t) = 0$ at $t = 0.405833 \dots$; $v_{\max} = 1.18 \text{ m s}^{-1}$.

3.3 The lowest point of v is $(4.712\dots, -e)$:
maximum speed $\boxed{e \approx 2.72}$ m s⁻¹.

3.4 (a)(i) $\boxed{1.81}$ cm. **(ii)** The graph of s is
steepest at $t = 0$: $v(0) = \frac{4\pi}{3} = \boxed{4.19}$ cm s⁻¹.
(b)(i) $\boxed{-1.47}$ cm. **(ii)** $s(3.5) = \boxed{1.07}$ cm.

§ 4

4.1 $A = \frac{\sqrt{3}}{4}x^2$, $\frac{dA}{dt} = \frac{\sqrt{3}}{2}x \cdot 4 = \boxed{30}$ cm² s⁻¹ at
 $x = 5\sqrt{3}$.

4.2 $r = \left(\frac{15}{\pi}\right)^{1/3} = 1.68389\dots$ and $\frac{dr}{dt} = \frac{5}{4\pi r^2} =$
 $\boxed{0.140}$ cm s⁻¹.

4.3 $V_{\max} = 2\sqrt{3}\pi$; half is $\sqrt{3}\pi$, reached at
 $h = 1.1818\dots$ $\frac{dV}{dh} = \pi(h^2 + 1)$, so $\frac{dh}{dt} =$
 $\frac{0.4}{\pi(h^2+1)} = \boxed{0.0531}$ m s⁻¹.

4.4 $h = 4.20648\dots$, $\frac{dh}{dt} = \frac{2}{10\pi h - \pi h^2} = \boxed{0.0261}$
cm s⁻¹.

§ 5

5.1 $\frac{dx}{dt} = -50 \operatorname{cosec}^2 \theta \frac{d\theta}{dt}$ with $\cot \theta = \frac{1}{5}$: $\frac{dx}{dt} =$
 $-50 \cdot \frac{26}{25} \cdot (-0.1) = \boxed{5.2}$ m s⁻¹.

5.2 $x = 4.63047\dots$, $\frac{d\theta}{dx} = -0.0294199\dots$, so $\frac{dx}{dt} = -424.88\dots$ and the speed is $\boxed{425}$ km h^{-1} .

5.3 $\sin \theta = \frac{14}{15}$, $x^2 = 625 - 600 \cos \theta$, $\frac{dx}{dt} = \frac{600 \sin \theta}{2x} \cdot \left(-\frac{\pi}{60}\right) = \boxed{-0.724}$ cm per minute.

§ 6

6.1 $9 - 3x - 2x^2 = 0$ gives $x = \frac{3}{2}$ (reject -3), and $y_R = -\sqrt{9 - \frac{9}{4}} = \boxed{-\frac{3\sqrt{3}}{2}}$.

6.2 $\frac{dT}{d\theta} = 500 \sec \theta \tan \theta - \frac{1000}{3} \sec^2 \theta = 0$ gives $\boxed{\sin \theta = \frac{2}{3}}$, so $\tan \theta = \frac{2}{\sqrt{5}}$ and $PX = 160\sqrt{5}$.

6.3 (i) $\frac{d^2L}{d\alpha^2} = \frac{45}{4}\sqrt{5} > 0$: concave $\boxed{\text{up}}$, so with $\frac{dL}{d\alpha} = 0$ it is a minimum. **(ii)** $L_{\min} = \frac{3}{4}\sqrt{5} + 6 \cdot \frac{\sqrt{5}}{2} = \boxed{\frac{15\sqrt{5}}{4}}$.

6.4 $11.25 = \frac{45}{4} > \frac{15\sqrt{5}}{4} \approx 8.39$: $\boxed{\text{no}}$.

6.5 $\boxed{\text{a}}$ — the largest value on a closed interval is at a vertex or an end; the ends give 0 and $V(x_m) > 0$.

§ 7

7.1 Minimise $l^2 = x^2 \ln x + 4$: $2x \ln x + x = 0$,
 $x > 0$, $\boxed{x = e^{-1/2}}$.

7.2 $D^2 = 136t^2 - 492t + 446$, least at $t = \frac{123}{68}$:
 $D_{\min} = \frac{\sqrt{1190}}{34} = \boxed{1.01}$ km.

7.3 $\left| \vec{AB} \times \vec{AC} \right|^2 = p^4 - 4p^3 + 14p^2 - 8p + 8$,
least at $p = 0.3264 \dots$: $\boxed{6.75}$.

§ 8

8.1 Growth is fastest at $P = \frac{k}{2}$: $t = 5 \ln \frac{k-1200}{1200} = \boxed{1.58}$ days.

8.2 (i) The minimum of f' (or the zero of f''): $e^x = 3 + 2\sqrt{2}$, $\boxed{x = 1.76}$. **(ii)**
 $f(1.76 \dots) = \sqrt{\frac{85}{\pi}} = \boxed{5.20}$ cm.

8.3 $kP \left(1 - \frac{P}{N}\right)$ has its vertex at $\boxed{P = \frac{N}{2}}$,
where $\frac{d^2P}{dt^2}$ changes from positive to negative.

8.4 $k \cdot \frac{N}{2} \cdot \frac{1}{2} = \boxed{\frac{kN}{4}}$.

§ 9

9.1 12, 36, 64, 81, 79.6, 64: $\boxed{P(12) = 81, n = 4}$.

9.2 $(20/7)^7 = 1554.26$ beats $(20/8)^8 = 1525.88$
: $\boxed{P(20) = 1550, n = 7}$.

9.3 $(100/37)^{37} = 9.474 \times 10^{15}$ beats
 $(100/36)^{36} = 9.400 \times 10^{15}$: $\boxed{9.47 \times 10^{15}}$.